

Introduction to Machine Learning Algo.

AI vs ML vs DL vs DS

Python developer

✓ AI Engineer

✓ Data Scientist.

Data Analyst.

EDA

visualizatⁿ AI

Data Scientist

(AI) Engineer

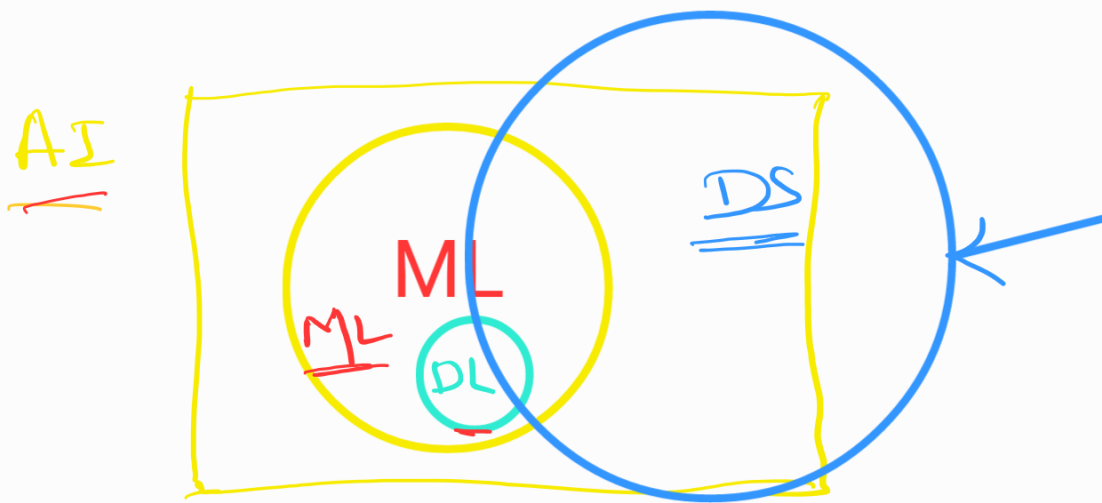
Data (Analyst)

(Big) data

(AI) engineer *knows*

DevOps engg → Docker

AI / ML / DL / DS



AI → An autonomous service

- ① Tesla → Self driving Car AI
↑↑
- ② YT → AI & DS → Recommⁿ ↓
engine
Netflix, jio, Sony, Instagram
Search → Facebook → SEO
- NLP

ML → 1. Statistical tools

2. Analysis of data

3. forecasting

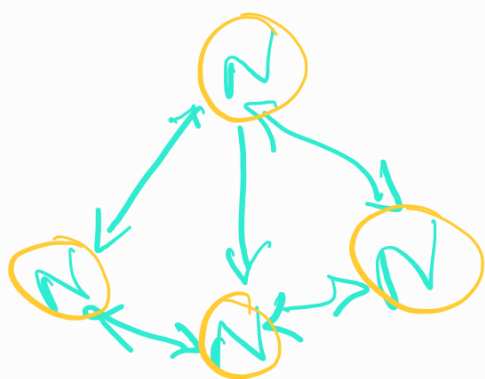
4. prediction



2024

2025

DL → Study of mimicing
Human Neuron



Patterns
Neuron ←

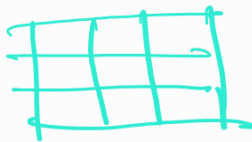
Human
Brain

- ① Single layer Neuro
- ② Multi level
- ③ RNN
- ④ CNN
- ⑤ ANN

Machine Learning & Deep learning

- ① Supervised ML → classification
→ Regression
- ② Unsupervised ML → clustering

① Supervised ML

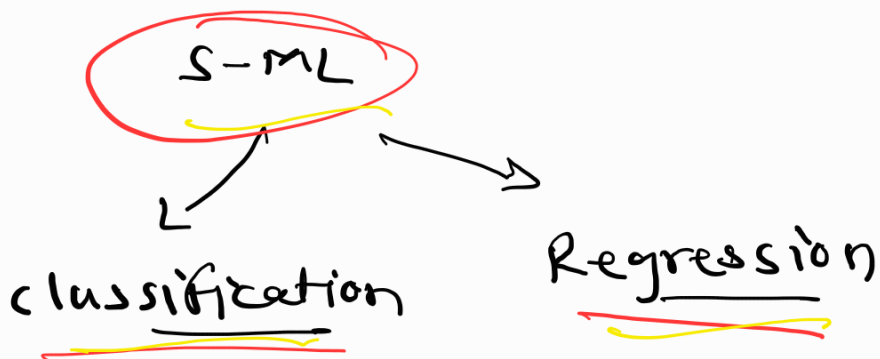
Dataset 

IFP features

Dependent feature

→	<u>SKIN_type</u>	✓ <u>eye_color</u>	✓ <u>weight</u>	✓ <u>Age</u>
	Wrinkled	blue	63	79
→	tight	Brown	89	41
	Damage	Black	23	18

→ Wrinkle Black 44 → 41 Age



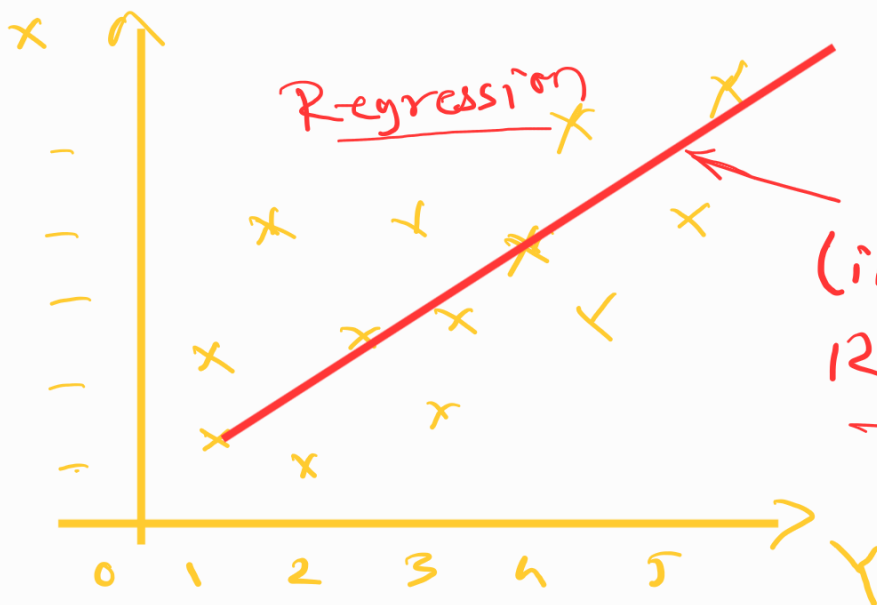
Ⓐ Classification → 2 prediction

<u>temp.</u>	<u>Humidity</u>	<u>Wind</u> <u>flow</u>	<u>dependent</u> <u>Rain/Not</u>
23°C	—	—	<u>Yes</u>
24°C	—	—	<u>No</u> 1
28°C	—	—	<u>Pass</u> 0
			<u>fail</u>

Ⓑ Regression : → Continuous variable

House Price Prediction

No. of Rooms Built up Area Carpet Area House price



1 cr
1.2 cr
5 cr

ML Algorithms

① Regression

1. Linear Regg.
2. Lasso & Ridge (L1/L2)
3. Decision tree regg.
4. Random forest regg.
5. XGBoost.
6. ANN
7. RNN

② Classification

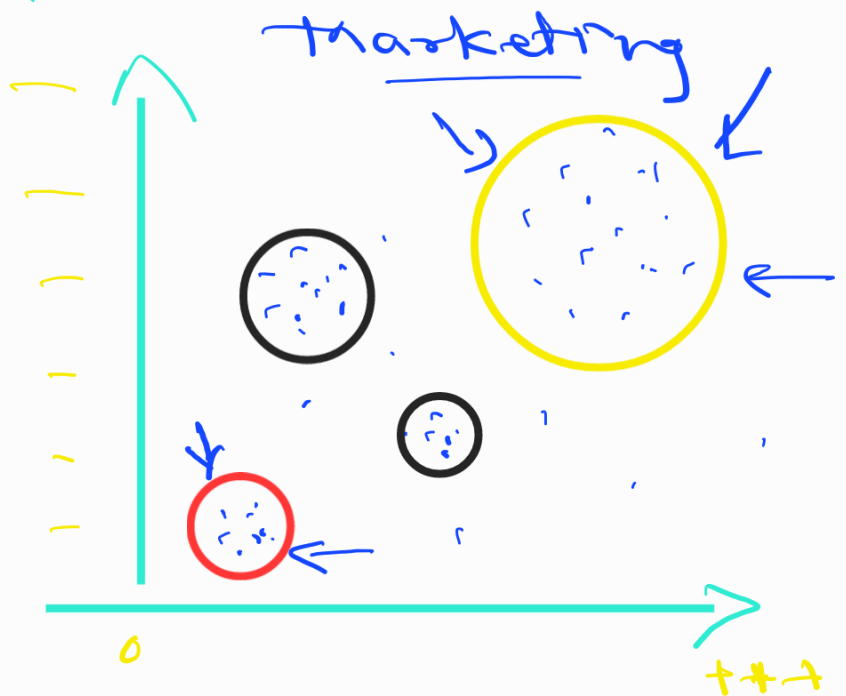
- ① logistic
- ② Decision tree
- ③ Random forest
- ④ XGboost
- ⑤ ANNA RNN

Unsupervised ML

① clustering

Sales data

- ① metro city
- ② tier - 2/3
- ③ Villager.



Time Series

→ ARIMA

